🧠 Neurological Physiotherapy — OSCE Preparation

Cases 5 & 6 · Parkinson's Disease — Early Stage & Advanced Stage

Target: Very Good (70%+) · Clinical Reasoning Throughout

Case 5 — Mr. Vance (Early PD) | Subjective | Pathophysiology | Differentials | Objective | Treatment | Case 6 — Mr. Miller (Advanced PD)

Case 5 — Mr. Leonard Vance

66 years old · Male | Parkinson's Disease · Diagnosed 12 months ago | Hoehn & Yahr Stage I–II (Early) | Resting tremor R hand · Micrographia · Hypokinetic gait

1 SUBJECTIVE ASSESSMENT

CONSENT / OPENING STATEMENT

"Good morning Mr. Vance, my name is [Name] and I'm the physiotherapist who will be working with you today. Before we begin, I'd like to explain that this session will involve me asking you some questions about your health, your day-to-day life, and what's been most difficult lately — and then I'll carry out some physical tests. Everything is voluntary and you can stop at any time. Are you happy to proceed? Is it okay if your wife stays in the room?" I make this

welcoming because in PD, hypomimia and bradyphrenia can make patients feel they are not communicating clearly — I want him to feel unhurried and respected from the outset.

KEY SUBJECTIVE QUESTIONS + CLINICAL REASONING

- **1. "What are the main things that have been bothering you most in your day-to-day life?"**

> **Why I ask this first:** I start patient-centred to identify his own priorities, not just textbook PD symptoms. In early PD, the patient's chief concern may be the tremor embarrassing him socially, difficulty writing cheques, or fear of future progression — not the gait slowness his wife notices. His goals drive my treatment priorities and improve adherence.

- **2. "Can you describe your walking — has it changed? Your wife mentioned you trail behind her; can you tell me more about that?"**

> **Why:** "Trailing behind" is a direct report of hypokinetic gait — reduced speed and step length from basal ganglia motor scaling failure. I want to know whether he also notices difficulty starting, freezing at doorways, or shuffling. This helps me distinguish between general bradykinesia (a dopaminergic issue) and true Freezing of Gait (FOG), which has a different management approach even in early PD.

- **3. "Tell me about getting out of a chair — when it's difficult, what exactly happens? Does it feel like your body won't start moving?"**

> **Why:** "Getting started" from a chair is akinesia — a failure of motor programme initiation. This is distinct from weakness or pain. Understanding this guides my treatment: I'll use cueing strategies (auditory "1-2-3-stand!") and LSVT BIG principles rather than strength training as the primary approach, because the problem is neurological initiation, not muscle capacity.

- **4. "How are your medications working? Do you notice times of day when you feel more in control of your movement — or times when it feels harder?"**

> **Why:** Even in early PD, levodopa timing creates "on" (well-medicated) and relative "off" periods. Identifying these windows is critical — I should schedule physiotherapy and home exercise during optimal "on" periods. It also establishes a baseline; when "off" periods worsen over time, this flags advancing disease and the need for neurologist medication review.

- **5. "Has your handwriting changed? Can you show me how you'd sign your name or write a sentence?"**

> **Why:** Micrographia is documented and directly reflects bradykinesia of the right upper limb. Asking him to demonstrate it in the room gives me an immediate functional snapshot of fine motor amplitude and speed — and I can use it as a subjective-to-objective bridge. It also identifies ADL impacts (bills, forms, correspondence) that may need OT referral.

- **6. "How has your mood been lately? And how are you and your wife managing at home — emotionally, not just physically?"**

> **Why:** Depression affects 40–50% of PD patients and is not merely reactive — it is part of the disease's non-motor profile from limbic dopamine and serotonin disruption. It also predicts worse physiotherapy outcomes if untreated. Asking about the wife separately screens for carer burden early, before it reaches crisis. I frame it gently because patients often minimise mood in a medical setting.

> **🚩 RED FLAGS — NAME, SIGNIFICANCE, AND ACTION**
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> **1. Sudden or rapid cognitive decline**
> Significance: May indicate Parkinson's Disease Dementia (PDD), Lewy Body Dementia, or medication toxicity. Atypically rapid onset suggests misdiagnosis — possible Parkinson's-Plus syndrome.

Action: Suspend physical assessment; complete MoCA cognitive screen; notify GP/neurologist urgently; do not proceed with exercise unsupervised.

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2. Severe orthostatic hypotension or syncopal episodes
Significance: Autonomic dysfunction in PD — a drop in systolic BP >20 mmHg on standing increases falls risk dramatically and can mimic worsening gait.
Action: Measure lying-to-standing BP before any mobility work; cease exercise if symptomatic; refer to neurologist for medication review (levodopa and dopamine agonists can worsen hypotension).

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3. Dysphagia or aspiration symptoms (coughing after eating/drinking)
Significance: Bulbar involvement affects swallowing safety; aspiration pneumonia is a leading cause of mortality in PD. Even in early stages this warrants urgent attention.
Action: Refer urgently to Speech and Language Therapy; advise patient and wife about safe swallowing positions; do not dismiss as minor.

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4. Unexpectedly early or severe postural instability with falls
Significance: Postural instability appearing very early in disease course is atypical for idiopathic PD and raises concern for PSP or MSA — Parkinson's-Plus syndromes with worse prognosis and poor levodopa response.
Action: Document trajectory carefully; flag for neurologist review at next appointment; adjust prognosis and treatment expectations accordingly.

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5. Visual hallucinations or paranoid delusions
Significance: Can be a dopaminergic medication side effect or early indicator of Lewy Body Dementia. Indicates CNS involvement beyond the motor system.
Action: Document exactly what the patient reports; do not

reinforce the content of hallucinations; refer to neurologist for medication review; inform carer.

PROBLEM LIST (ICF FRAMEWORK)

PHYSICAL IMPAIRMENTS

- Resting tremor — right hand (4–6 Hz)
- Bradykinesia — slowed, reduced-amplitude movement
- Akinesia — difficulty initiating movement (chair)
- Reduced right arm swing during gait
- Stooped posture — thoracic kyphosis, flexor dominance
- Decreased bilateral hip and knee flexor ROM
- Micrographia — fine motor hypokinesia
- Likely cogwheel/leadpipe rigidity (to confirm objectively)

FUNCTIONAL ACTIVITY LIMITATIONS

- Reduced walking speed — cannot keep pace with wife
- Difficulty initiating sit-to-stand from low chairs
- Impaired handwriting — impacts ADLs (correspondence, forms)
- Reduced participation in social/community walking
- Potential driving safety concern — to explore with patient
- Possible reduced household task completion

PSYCHOSOCIAL / YELLOW FLAGS

- **Fear of progression:** Early PD diagnosis can generate anticipatory anxiety about future disability; this can cause premature activity avoidance and deconditioning
- **Social embarrassment:** Visible tremor and slow walking in public may cause the patient to withdraw from activities, worsening

isolation and mood

- **Carer concern and emerging role shift:** Wife noticing gait change suggests she is monitoring him — this is the beginning of a carer role that needs to be managed proactively to avoid over-protection and learned helplessness
- **Depression/anxiety:** Screen with GDS-15 or HADS — non-motor feature of PD, not just a reaction to diagnosis; requires MDT attention if positive
- **Occupational identity:** If Mr. Vance was previously active or employed, loss of physical capability threatens his sense of self — explore what he valued doing before symptoms emerged

2 PATHOPHYSIOLOGY OF WORKING DIAGNOSIS

WHAT IS HAPPENING NEUROLOGICALLY

Parkinson's Disease is a progressive neurodegenerative disorder characterised by the selective loss of dopaminergic neurons in the substantia nigra pars compacta (SNc). These neurons project via the nigrostriatal pathway to the striatum (caudate nucleus and putamen), where dopamine normally facilitates the direct pathway — the "go" signal for voluntary movement — while modulating the indirect pathway — the "stop" signal that suppresses unwanted motor programmes. As dopaminergic neurons are lost (typically 60–80% must be lost before motor symptoms appear), the direct pathway becomes underactive and the indirect pathway becomes overactive, resulting in excessive inhibitory output from the globus pallidus interna (GPi) and substantia nigra pars reticulata (SNr) to the thalamus.

This thalamic suppression dampens motor cortex excitation, producing the cardinal features of hypokinesia, bradykinesia, and akinesia. Concurrently, misfolded alpha-synuclein aggregates form Lewy bodies within surviving neurons; these propagate through the nervous system in a predictable pattern (Braak staging), eventually involving the cortex,

limbic system, and autonomic nervous system — explaining the wide spectrum of non-motor symptoms. By the time of Mr. Vance's diagnosis, the disease process was likely well established for years before his resting tremor became the presenting complaint.

TYPICAL SIGNS AND SYMPTOMS — EXPLAINED BY PATHOPHYSIOLOGY

- **Resting tremor (4–6 Hz, "pill-rolling"):** Arises from pathological oscillations in the thalamo-cortical and basal ganglia-thalamic circuits when normal dampening by the dopaminergic system is lost. Suppressed during voluntary movement because cortical volitional drive temporarily overrides the oscillation — this is why the tremor disappears when he reaches for a cup.
- **Bradykinesia and akinesia:** Direct consequence of reduced dopaminergic facilitation through the direct pathway — the motor cortex receives less thalamic excitation, meaning movement initiation requires much greater cortical effort. Movement amplitude and velocity progressively decline, and initiation is delayed or fails (akinesia).
- **Rigidity (cogwheel or leadpipe):** Loss of basal ganglia modulation of descending motor pathways leads to co-contraction of agonist and antagonist muscle groups simultaneously. Passive joint movement feels uniformly stiff (leadpipe) or intermittently "catching" (cogwheel, thought to be rigidity superimposed on tremor).
- **Stooped posture and reduced arm swing:** Flexor dominance emerges from rigidity — the flexor muscle groups are tonically over-active relative to extensors. Reduced arm swing is a direct expression of right-sided bradykinesia and rigidity reducing the amplitude of the normal reciprocal arm movement during gait.
- **Micrographia:** A form of bradykinesia specific to sequential, repetitive fine motor tasks — movement amplitude progressively decreases during the task as dopaminergic drive is insufficient to sustain normal motor scaling across repeated strokes.

WHY THIS PATIENT FITS THE PATHOPHYSIOLOGY

Mr. Vance presents with three cardinal motor features of idiopathic PD — resting tremor (right hand), bradykinesia (micrographia, slow gait, chair initiation difficulty), and postural changes (stooped posture, reduced arm swing) — with a classic asymmetric onset affecting the right side preferentially. This lateralised presentation is the hallmark of idiopathic PD, reflecting greater dopaminergic neuron loss in the left SNc. His difficulty initiating sit-to-stand is akinesia — the basal ganglia's failure to trigger the motor programme — not weakness. At 12 months post-diagnosis and Hoehn & Yahr I–II, he retains functional independence, consistent with early-stage disease where compensatory cortical mechanisms can still partially offset basal ganglia dysfunction.

3 DIFFERENTIAL DIAGNOSES

DIFFERENTIAL 1: ESSENTIAL TREMOR (ET)

- Most common movement disorder in adults; bilateral upper limb tremor is frequent in men over 60
- Supporting features that could overlap: upper limb tremor, older male, affecting handwriting
- Key distinguishing features of ET: tremor is *kinetic and postural* (worsens on action and sustained posture, e.g. holding a cup outstretched) — opposite to PD resting tremor which *reduces* on action; no bradykinesia; no rigidity; no gait or posture changes; often positive family history; improves with alcohol; no response to levodopa

DIFFERENTIAL 2: PROGRESSIVE SUPRANUCLEAR PALSY (PSP)

- A Parkinson's-Plus syndrome with parkinsonism, early falls, and supranuclear gaze palsy

- Supporting features that could overlap: stooped posture (though PSP tends to cause axial extension rather than flexion), bradykinesia, gait difficulty, older male
- Key distinguishing features of PSP: *axial rigidity greater than limb rigidity*; early severe postural instability and falls (within first year); vertical gaze palsy (inability to look down — patients cannot cut food or read); poor or absent levodopa response; more rapid decline; symmetric rather than asymmetric onset

Clinical Reasoning — Why I Favour Idiopathic PD:
I favour idiopathic Parkinson's Disease over Essential Tremor because Mr. Vance's tremor is explicitly a *resting* tremor — the most specific indicator of basal ganglia dysfunction — accompanied by bradykinesia (micrographia, slow gait, akinetic sit-to-stand) and postural changes, none of which occur in ET. I favour PD over PSP because his onset is insidious over 12 months with predominantly right-sided (asymmetric) features, he has no early severe falls or reported gaze difficulty, and his posture is flexed rather than extended. PSP also typically progresses much faster and causes falls within the first year. The confirmed neurologist diagnosis further supports idiopathic PD as the working diagnosis.

4 OBJECTIVE ASSESSMENT — SPECIAL TESTS

KEY NEUROLOGICAL OBSERVATIONS (BEFORE FORMAL TESTING)

- **Tone:** Passive movement of wrist and elbow bilaterally — assess for cogwheel (intermittent catches) or leadpipe (uniform) rigidity; note asymmetry (right worse expected); Froment's manoeuvre (asking patient to swing contralateral arm) can enhance subtle rigidity
- **Coordination:** Finger-nose test (absence of dysmetria distinguishes from cerebellar pathology); rapid alternating movements — pronation/supination (dysdiadochokinesia may be present due to

bradykinesia, but should not have the irregular timing of cerebellar dysfunction)

- **Posture:** Observe standing — degree of thoracic kyphosis, forward head posture, flexed hip/knee stance; measure camptocormia (trunk flexion angle) if severe
- **Gait observation:** From front, side, and behind — step length, symmetry of arm swing, heel strike, cadence, turning strategy, festination; note any freezing at doorway on entry
- **Facial expression:** Hypomimia (masked facies) — reduced spontaneous facial movement; note blink rate (reduced in PD)
- **Speech:** Volume (hypophonia), pitch variation (monotone), rate; note for SALT referral if significant

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Test 1: Timed Up and Go (TUG)

- **What it measures:** A composite measure of functional mobility including sit-to-stand, walking, turning (180°), walking back, and returning to sitting. It captures gait initiation speed, walking velocity, turning ability, and sit-to-stand — exactly the functional problems Mr. Vance reports. It also serves as a fall risk screening tool.
- **How to perform:**
  1. Use a standard chair with armrests (height ~46 cm); patient seated fully back in the chair
  2. Place a cone or mark on the floor exactly 3 metres in front of the chair
  3. Instruct: "When I say go, stand up, walk to the marker, turn around, walk back, and sit down. Walk at your comfortable, normal pace."
  4. Begin timing on "go"; stop the clock when the patient is fully seated
  5. Allow one practice trial to familiarise; record two timed trials and take the mean

6. Observe quality: hesitation at initiation, arm swing, step length, turning strategy, how they lower to sit

- **Positive / Abnormal finding:** >12 seconds is associated with increased fall risk in community-dwelling older adults. In PD, >11.5 seconds indicates significant mobility impairment. However, I pay as much attention to *quality* as time — hesitation at sit-to-stand, a narrow shuffling turn, absent arm swing, and a slow lower back to the chair are all clinically significant observations even if time is borderline.
- **Why I chose it:** TUG directly captures both of the functional concerns raised — difficulty starting from a chair and reduced walking speed while also providing a standardised, repeatable outcome measure. It requires no equipment beyond a chair and a stopwatch, can be performed in any clinic room, and has good sensitivity for detecting early gait impairment in PD. It is my primary baseline measure for tracking treatment response.

Test 2: Pull Test (Postural Instability — MDS-UPDRS Item 3.12)

- **What it measures:** Integrity of automatic postural righting reflexes — the ability to recover from an unexpected backward perturbation without taking excessive corrective steps or falling. Directly assesses a key domain of the Hoehn & Yahr staging (postural instability distinguishes H&Y III from II).
- **How to perform:**
  1. Patient stands with feet shoulder-width apart, eyes open, in a clear space
  2. Inform patient: "I'm going to give you a firm pull backwards on your shoulders. Try to stay balanced — you may take a step if you need to."
  3. Stand directly behind the patient, hands on both shoulders; position yourself to catch them

4. Deliver a brisk, firm backward pull (not a light nudge — it should be genuinely challenging)
5. Count the number of corrective steps taken; observe whether they recover independently

- **Positive / Abnormal finding:** Normal = 0–1 corrective steps. Mildly abnormal = 2–3 steps (UPDRS score 1–2). Severely abnormal = 4+ steps, or would fall without being caught (UPDRS score 3–4). Even 2 corrective steps in early PD is clinically meaningful as it predicts future falls and indicates postural reflex impairment earlier than patients typically report symptoms.
- **Why I chose it:** Postural instability is the single strongest predictor of falls in PD and is the key variable that differentiates Hoehn & Yahr stages. Establishing this baseline at 12 months — when it may still be mildly impaired or normal — allows me to detect deterioration at future reviews, justify proactive balance training even before instability becomes severe, and communicate clearly to the neurologist about staging.

Test 3: Finger-Tapping Speed & Amplitude (MDS-UPDRS Item 3.4)

- **What it measures:** Bradykinesia of the right upper limb — specifically the speed, amplitude, and pattern of repetitive sequential finger movements. A decremental pattern (progressively slowing and decreasing amplitude over 10 seconds) is highly characteristic of PD and reflects dopaminergic motor drive failure during sustained motor output.
- **How to perform:**
  1. Demonstrate to patient: tap index finger to thumb repeatedly, as fast as possible and as wide as possible — emphasise "big and fast"

2. Ask patient to mirror this movement; test non-dominant (left) hand first as a baseline, then dominant right (affected) hand
3. Each hand performs 10 seconds of continuous tapping
4. Observe and score: speed, amplitude, regularity, and whether amplitude or speed decreases across the 10 seconds (decremental pattern)

- **Positive / Abnormal finding:** Progressive decrease in amplitude or speed over the 10 seconds, marked right-left asymmetry (right worse), or hesitations mid-task. This confirms upper limb bradykinesia and quantifies the severity — directly linking the objective finding to the subjective micrographia complaint. A clear asymmetry (right worse than left) is consistent with left SNc predominance in idiopathic PD.
- **Why I chose it:** Mr. Vance has documented micrographia and a right resting tremor. I need an objective, standardised measure of right upper limb motor function — this test is embedded in the MDS-UPDRS (the gold standard PD motor scale), so my findings are directly comparable across review points and communicable to the neurology team. It also directly informs my upper limb exercise prescription and any referral to OT for handwriting rehabilitation.

Outcome Measures

MDS-UPDRS Part III (Motor Examination): Gold standard for quantifying PD motor impairment across tremor, rigidity, bradykinesia, gait, and postural stability domains. Administer at baseline and every 3 months to track progression and treatment response. Score 0–132 (higher = more impaired).

MiniBESTest (Mini Balance Evaluation Systems Test): 14-item balance assessment covering anticipatory postural adjustments, reactive postural control, sensory orientation, and dynamic gait. Score 0–28. More sensitive than the Berg Balance Scale for detecting

early balance deficits in PD — appropriate for Hoehn & Yahr I–II where deficits are subtle. A score of <20 indicates moderate balance impairment requiring intervention.

5 TREATMENT PLAN

INITIAL PHASE — WEEKS 1–4: RATIONALE

My overarching goals in this phase are to address bradykinesia and hypokinesia through high-amplitude, high-effort movement training (consistent with LSVT BIG principles), improve gait quality and sit-to-stand function, and build Mr. Vance's and his wife's self-management capability. Evidence strongly supports that physiotherapy initiated in early PD — before compensatory strategies become entrenched — exploits remaining neuroplasticity to slow functional decline. I will schedule sessions during his documented "on" medication window.

Exercise 1: LSVT BIG — Large Amplitude Whole-Body Movements

Clinical reasoning: In PD, the internal sense of "normal" movement amplitude is recalibrated downward by the basal ganglia — Mr. Vance's brain believes small movements are normal-sized. LSVT BIG exploits preserved cortical pathways by training conscious, exaggerated large-amplitude movements. The key insight is that what feels "too big" to the patient is actually closer to normal. Evidence demonstrates improvements in gait speed, stride length, balance, and ADLs following this approach.

How to perform: Select 8 whole-body functional movements (e.g. large arm swings forward and back, wide step forward and return, trunk rotation left and right, overhead reach). Instruct with maximum amplitude: "Think BIG — make every movement as large as you possibly can." Use verbal cueing: "BIG step," "WIDE

swing." Perform 8 repetitions × 4 sets, daily. Apply the principle to ADLs immediately: "Take a BIG step when you stand up. Swing your arms BIG when you walk."

- **Regression:** Seated position; reduce to 4 repetitions; therapist provides tactile cueing on limb to guide amplitude; focus on single movements before combining; reduce pace
- **Progression:** Add dual-task challenge (count backwards by 3 while performing big steps); increase walking speed while maintaining big arm swing; practise on outdoor uneven terrain; carry light objects while walking; introduce to group exercise setting for social reinforcement

Exercise 2: Cueing-Based Sit-to-Stand Training

Clinical reasoning: Sit-to-stand difficulty is akinesia — failure to initiate a stored motor programme. External cueing strategies (auditory counts, visual targets, attentional focus instructions) activate the supplementary motor cortex and lateral premotor cortex directly, bypassing the deficient basal ganglia initiation circuit. This is one of the most functionally critical transfers in daily life and a documented safety risk (falling when rising from a chair).

How to perform: Seat Mr. Vance at edge of chair, feet hip-width apart behind knees, hands on thighs. Teach the "nose-over-toes" strategy: lean trunk forward until nose is over toes before standing — this shifts the centre of mass forward over the feet, reducing the motor threshold for initiation. Apply rhythmic auditory cue: therapist counts "1... 2... 3... STAND!" or patient learns to self-cue. Also practise using a floor target to step onto after standing (provides a movement goal to drive initiation). 10 repetitions × 3 sets per session.

- **Regression:** Raise chair seat height (use cushion or raised toilet frame); allow armrest push-up; reduce to 5 repetitions;

therapist provides light forward assist at pelvis

- **Progression:** Lower seat height progressively; remove armrests; add a forward step immediately on standing; introduce at varying times of day (including relative "off" periods with carer supervision); dual task (say a word beginning with each letter of the alphabet on each repetition)

SAFETY CONSIDERATIONS

- **Falls risk stratification:** Complete the Falls Risk Assessment Tool (FRAT); arrange OT home assessment for environmental hazards (rugs, lighting, bathroom rails)
- **Medication window scheduling:** All physiotherapy sessions and home exercise should be timed 30–60 minutes after morning levodopa dose — during optimal "on" period — when motor control is best and exercise is safest and most effective
- **Orthostatic hypotension precautions:** Instruct Mr. Vance to always pause on the edge of the chair before fully standing; monitor for dizziness; rest between exercise sets; adequate hydration before sessions
- **Fatigue management:** PD patients fatigue disproportionately — build rest intervals into every session; educate on activity pacing across the day; shorter more frequent exercise bouts are preferable to long exhausting sessions
- **Carer supervision:** Ensure wife is present for the first 2–3 home exercise sessions to observe correct technique, provide cueing, and ensure safety before Mr. Vance exercises independently

PATIENT AND CARER EDUCATION — KEY MESSAGES

- **1. "Think BIG, Move BIG":** "Your brain is underestimating how much you're moving. You need to consciously exaggerate every movement — bigger arm swings, longer steps, louder voice — to achieve what used to feel normal. This is the disease, not laziness or weakness. The strategy is real and evidence-based."

- **2. Cueing strategies for freezing and initiation difficulty:** "If you feel stuck, try counting aloud — '1-2-3-step' — or follow a line on the floor, or shift your weight from side to side. These tricks use a different part of your brain to get you moving again, one that is less affected by Parkinson's."
- **3. Exercise is neurological medicine:** "Research shows regular aerobic and amplitude-based exercise does not just maintain fitness — it may slow the progression of Parkinson's symptoms by supporting your brain's own repair and adaptation processes. Aim for at least 150 minutes of moderate exercise per week."
- **4. For the wife — how to help without over-helping:** "Walking beside Leonard — not behind him — helps him keep a good pace. Avoid rushing him; pressure actually worsens freezing. If he freezes, count aloud with him rather than pulling him. Let him do as much as he safely can independently — preserving his capability and confidence is as important as physical safety."

PROGRESSION CRITERIA (MEASURABLE TARGETS TO ADVANCE)

- TUG time improves by ≥1.5 seconds from baseline
- Sit-to-stand completed in under 3 seconds without armrests, independently, on 8/10 attempts
- MiniBESTest score improves by ≥3 points
- Patient independently applies LSVT BIG principles during gait without verbal prompting (wife confirms at home)
- No falls in the preceding 2 weeks

LONG-TERM PLAN — WEEKS 4–12+

- **Community exercise programme:** Refer to PD-specific group exercise (Boxing4Parkinson's, PD Warrior, Nordic Walking, Tai Chi, dance therapy) — these provide dual-task challenge, social engagement, motivation, and evidence-based motor benefit; Tai Chi specifically has strong evidence for improving balance in PD

- **MDT referrals:** Occupational therapy (full ADL and home assessment, driving review, handwriting aids); Speech and Language Therapy if hypophonia or dysphagia develops; neuropsychology if depression/anxiety confirmed on screening
- **Self-management:** Provide Parkinson's UK/Australia patient resources; link to local PD support group; establish exercise diary; NeurOptimal or gamified exercise if available to improve adherence
- **Review schedule:** Physiotherapy review every 3 months; re-administer MDS-UPDRS Part III and MiniBESTest at each review; escalate to neurologist if motor fluctuations develop or Hoehn & Yahr stage advances

☛ Examiner Demonstration — What I Can Show In the Room

Technique 1 — Auditory Cueing for Gait Initiation: Ask the examiner to walk as the patient. Stand beside them and use a rhythmic verbal count — "1-2-1-2" — or clap a steady beat. Demonstrate how altering the beat cadence modulates step rate. Explain: "Auditory cueing drives the supplementary motor cortex and cerebellum directly via the auditory-motor coupling network, bypassing the basal ganglia timing circuit that is disrupted in PD. This is why something as simple as counting aloud can transform a shuffling patient's gait."

Technique 2 — LSVT BIG Arm Swing Demonstration: Demonstrate the key BIG exercise — wide, exaggerated pendular arm swings from the shoulder while stepping in place, with deliberate over-amplitude. Ask the examiner to try it. Instruct: "Make it feel ridiculously big — that's the point." Explain: "The brain's internal sense of normal amplitude has been recalibrated downward by dopamine loss. By training consciously at what feels exaggerated, we recalibrate the patient's movement towards normal through cortical pathways, rather than relying on the deficient subcortical basal ganglia system."

Case 6 — Mr. George Miller

75 years old · Male | Parkinson's Disease · 12 years duration | Hoehn & Yahr Stage IV (Advanced) | Motor fluctuations · Two falls (turning) · Shuffling gait · FOG

1 SUBJECTIVE ASSESSMENT

CONSENT / OPENING STATEMENT

"Good morning Mr. Miller, my name is [Name] and I'm your physiotherapist today. I know you've had some difficult falls recently, so before anything else I want you to know that your safety is our top priority today — we'll take everything at your pace and I won't ask you to do anything that feels unsafe. I'll start by asking you some questions, and then we'll have a look at how you're moving. If you need to rest, just say the word. Are you comfortable to begin? Is there anyone with you today — a family member or carer — who I can also speak with?" I take extra care in this consent because fear of falling after two recent falls may make Mr. Miller anxious about assessment; a calm, unhurried opening reduces that anxiety and improves the reliability of my objective findings.

KEY SUBJECTIVE QUESTIONS + CLINICAL REASONING

- **1. "Tell me exactly what happened in each of the two falls last month — where were you, what were you doing, and what happened in the seconds just before you went down?"**

> **Why this is my very first question:** Falls analysis is the highest clinical priority at this presentation. Both falls occurred during turning in the kitchen — this is a specific and critical pattern that points to Freezing of Gait (FOG) triggered by a confined, high-cognitive-

demand environment with directional changes. Understanding the exact mechanism (FOG-induced fall vs. postural instability vs. orthostatic hypotension vs. near-syncope) determines my entire intervention strategy. If both falls were FOG, cueing training and wide turning strategies are my primary tools. If they were balance-related without FOG, I focus on reactive postural control training instead.

- **2. "Describe a freezing episode to me — what does it feel like? Can you predict when it's going to happen? Does it happen at doorways, when you're turning, or when you're in a hurry?"**

> **Why:** Mapping the specific triggers and context of FOG is essential for targeted intervention. FOG triggers include: narrow spaces, doorways, turning (especially in confined spaces), dual-tasking, destination anxiety, and "off" medication periods. Mr. Miller's kitchen falls tell me confined-space turning is a trigger — but I need to know the full trigger profile to design safe home recommendations, cueing strategies, and walking aid prescription. Some aids (standard rollators) worsen FOG; others (laser-line walkers) are highly effective.

- **3. "Walk me through a typical 'off' period — how do you know it's starting, how long does it last, and what can you do during one? Are they predictable or do they hit suddenly?"**

> **Why:** After 12 years of PD and levodopa therapy, motor fluctuations are expected — surviving neurons can no longer buffer dopamine smoothly, so blood levodopa levels oscillate dramatically. Distinguishing predictable wearing-off from sudden on-off fluctuations matters clinically: wearing-off follows dose timing and can be managed by adjusting the schedule; sudden on-off is less predictable and represents a much higher falls risk. I also need to know what Mr. Miller can and cannot do safely during "off" periods — this determines whether he should be attempting any mobility unsupervised.

- **4. "What walking aids do you currently use — and do they help, or do you feel they ever get in the way or make freezing worse?"**

Why: This is a prescription-critical question in advanced PD. A standard 4-wheeled rollator with a large front bar can act as a visual obstacle that worsens FOG — the patient freezes behind their own walking frame. Conversely, a U-Step walker with an embedded laser line projects a visual target on the floor that actively breaks FOG. If Mr. Miller is using the wrong aid, changing it may immediately reduce his falls risk more than any exercise. I cannot assume his current aid is appropriate without asking.

- **5. "How is your home set up — are there any steps, rugs, or narrow spaces? Who else is at home with you, and are they around when you need them most?"**

Why: At Hoehn & Yahr IV, Mr. Miller is severely disabled. Both his falls were at home in the kitchen — the environment is an active risk factor. I need to know his living layout, what environmental hazards exist, and what his support network looks like. If he is ever alone at home during "off" periods, this is a patient safety emergency that requires immediate action — not just a note to follow up later.

- **6. "Do you have any chest pain, breathlessness, or dizziness when you're moving around — particularly when you first stand up?"**

Why: At 75 with advanced PD, cardiovascular comorbidities and autonomic dysfunction are both likely. Orthostatic hypotension (systolic BP drop >20 mmHg on standing) is very common in advanced PD due to autonomic neuron involvement and is both a falls risk and a barrier to exercise. Cardiac disease would require modification of my exercise prescription and potentially a medical clearance before I commence aerobic training. I need to rule these out before I proceed to objective testing involving standing and walking.

🚩 RED FLAGS — NAME, SIGNIFICANCE, AND ACTION

1. Recurrent injurious falls — already present

Significance: Two falls last month is an active falls crisis. At Hoehn & Yahr IV with impaired postural reflexes, falls carry a high risk of hip fracture, traumatic brain injury, and prolonged floor time — all associated with significant morbidity and mortality in advanced PD.

Action: Immediate multifactorial falls risk assessment; expedite OT home visit; prescribe personal emergency alarm pendant today; assess bone density (refer to GP); ensure safe supervision arrangements are in place before patient is left alone.

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2. Severe orthostatic hypotension or syncope

Significance: A BP drop on standing causing near-syncope or collapse can mimic or compound FOG-related falls. Autonomic failure is common in advanced PD and is worsened by levodopa, dopamine agonists, and dehydration.

Action: Measure lying and standing BP immediately before any mobility-based assessment; if symptomatic, cease all standing activity; refer to neurologist for medication review; advise adequate hydration and compression stockings.

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3. Signs of aspiration or significant dysphagia

Significance: Bulbar dysfunction in advanced PD impairs swallowing coordination; aspiration pneumonia is the leading cause of death in advanced PD. Reduced cough reflex may mean aspiration is silent.

Action: Screen with a water swallow test if indicated; refer urgently to Speech and Language Therapy; advise on safe swallowing posture; consider texture modification; escalate to medical team if respiratory symptoms are present.

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4. Acute cognitive deterioration, confusion, or delirium

Significance: PD dementia affects up to 80% of patients at 20 years. Acute confusion may indicate infection (UTI, pneumonia), medication toxicity, or delirium superimposed on existing cognitive impairment — all require urgent medical review.

Action: Suspend assessment; notify medical team; basic cognitive screen (abbreviated MMSE); check for signs of infection; do not attempt any exercise or mobility work until medically cleared.

5. Signs of respiratory compromise (reduced chest expansion, low SpO2, tachypnoea)
Significance: Thoracic rigidity, kyphosis, and potential aspiration reduce respiratory reserve in advanced PD; acute decompensation is life-threatening.
Action: Check oxygen saturation; respiratory physiotherapy assessment; position for optimal lung volumes; escalate to medical team and consider hospital admission if SpO2 <94%.

PROBLEM LIST (ICF FRAMEWORK)

PHYSICAL IMPAIRMENTS

- Freezing of gait (FOG) — turning-triggered, confirmed by falls context
- Shuffling gait — reduced step length, increased cadence, poor heel clearance
- On-off motor fluctuations — sudden loss of mobility unpredictably
- Severe postural instability (Hoehn & Yahr IV — expected)
- No ROM deficits — impairment is central/neurological, not musculoskeletal

FUNCTIONAL ACTIVITY LIMITATIONS

- High falls risk — two falls in past month, both at home
- Unable to safely turn in confined spaces (kitchen)
- Likely housebound or near-housebound — community mobility severely restricted
- Increasing ADL dependence during "off" periods
- Significant carer burden — 12-year disease, escalating care needs
- Safety risk if ever alone at home during "off" period

- Likely significant rigidity (advanced disease)
- Probable autonomic dysfunction (to confirm with lying/standing BP)
- Likely dual-task cost elevation (to quantify objectively)

PSYCHOSOCIAL / YELLOW FLAGS

- **Fear of falling (kinesiophobia):** Two recent falls will almost certainly have generated significant fear of further falls; this can lead to catastrophic activity restriction and rapid deconditioning — a vicious cycle in advanced PD
- **Carer burden:** 12 years of caregiving is exhausting; carer may be burnt out or developing their own health problems — I must assess carer capacity and refer for respite if needed
- **Loss of autonomy and identity:** The progression from walking independently to requiring supervision for basic mobility is a profound loss; depression in advanced PD is extremely common and significantly worsens prognosis
- **Health literacy around "on-off" management:** Does Mr. Miller understand the relationship between medication timing and safe mobility? Lack of understanding leads to dangerous unsupervised activity during "off" periods

2 PATHOPHYSIOLOGY OF WORKING DIAGNOSIS

WHAT IS HAPPENING NEUROLOGICALLY — ADVANCED PD

After 12 years, Mr. Miller's disease has progressed to a state of near-total dopaminergic neuron loss in the substantia nigra, with Lewy body pathology (Braak stage V–VI) extensively involving the frontal cortex,

limbic system, brainstem reticular formation, and autonomic nervous system. The basal ganglia thalamocortical motor circuits are profoundly disrupted — the GPi and SNr generate massive inhibitory output onto the thalamus, suppressing motor cortex activation to a degree that makes voluntary movement initiation extraordinarily effortful and unreliable, particularly for complex, multi-step motor sequences such as turning.

Freezing of Gait (FOG) in advanced PD is understood to arise from dysfunction at multiple levels: (1) failure of the mesencephalic locomotor region (MLR) and pedunculopontine nucleus (PPN) to receive adequate facilitation from the basal ganglia, disrupting the spinal locomotor rhythm generator; (2) impaired communication between the subthalamic nucleus and PPN that normally coordinates gait initiation and continuous stepping; and (3) attentional resource competition — when cognitive demands are high (navigating a small kitchen, planning a directional change), the deficient motor system cannot simultaneously process spatial navigation and maintain continuous gait rhythm, resulting in a complete cessation of movement. This explains why FOG is most dangerous in confined, complex environments.

Motor fluctuations — the "on-off" phenomenon — arise because after 12 years of levodopa therapy, the remaining dopaminergic neurons have lost their capacity to buffer and gradually release dopamine. Dopamine availability in the striatum now mirrors levodopa plasma levels directly, oscillating sharply with each dose cycle. When levodopa levels fall (wearing-off) or fluctuate unpredictably (on-off), the striatum loses dopaminergic tone abruptly, and the patient transitions from functional mobility to near-immobility within minutes. This sudden transition — combined with FOG and impaired postural reflexes — is what makes advanced PD so dangerous from a falls perspective.

WHY THIS PATIENT FITS THE PATHOPHYSIOLOGY

Mr. Miller's 12-year disease duration and Hoehn & Yahr IV status are consistent with the degree of dopaminergic depletion and Lewy body spread expected at this stage. His shuffling gait — reduced step length,

increased cadence — is the prototypical advanced PD hypokinetic gait pattern, in which the brain cannot generate adequate motor drive to produce normal step length, and instead compensates with rapid, short steps to maintain forward momentum. Critically, the absence of ROM deficits on evaluation is neurophysiologically consistent with advanced PD: the joints themselves are mobile because the limitation is a central failure of motor programme scaling and generation, not a peripheral musculoskeletal restriction. His FOG triggered specifically by turning in a confined kitchen is a textbook example of the attentional-motor resource competition model of FOG.

3 DIFFERENTIAL DIAGNOSES

DIFFERENTIAL 1: MULTIPLE SYSTEM ATROPHY — PARKINSONIAN TYPE (MSA-P)

- A Parkinson's-Plus syndrome combining parkinsonism, cerebellar features, and severe autonomic failure
- Supporting features that could overlap: significant motor disability, falls, autonomic dysfunction, shuffling gait, motor rigidity
- Key distinguishing features of MSA: severe early autonomic failure (symptomatic orthostatic hypotension, urinary incontinence) as a prominent feature; cerebellar signs (ataxia, dysarthria, nystagmus); much more rapid progression — most patients with MSA reach severe disability within 3–5 years and median survival is 6–9 years from diagnosis, making a 12-year disease course highly inconsistent with MSA; poor levodopa response

DIFFERENTIAL 2: VASCULAR PARKINSONISM (VP)

- Parkinsonism caused by multiple small vessel cerebrovascular events (lacunar infarcts) affecting the basal ganglia and white matter circuits

- Supporting features that could overlap: shuffling gait, falls, lower limb greater than upper limb involvement, older age, male
- Key distinguishing features of VP: predominantly lower body parkinsonism (gait disorder disproportionate to upper limb bradykinesia); early prominent cognitive impairment; history of vascular risk factors (hypertension, diabetes, stroke); stepwise rather than gradual progression; no significant tremor; poor levodopa response; MRI shows white matter changes or lacunar infarcts

Clinical Reasoning — Why I Favour Advanced Idiopathic PD:
I favour idiopathic Parkinson's Disease over MSA-P because 12 years of disease duration is incompatible with MSA, which has a median survival of 6–9 years — Mr. Miller's longevity strongly argues against this diagnosis. Most decisively, the report of "on-off" motor fluctuations directly implies a robust levodopa response — both MSA and Vascular Parkinsonism characteristically fail to respond to levodopa, whereas idiopathic PD responds well, especially early in the disease course. The gradually progressive course, the documented asymmetric history typical of idiopathic PD, and the well-established long-term levodopa treatment all point firmly to advanced idiopathic PD as the correct working diagnosis.

4 OBJECTIVE ASSESSMENT — SPECIAL TESTS

KEY NEUROLOGICAL OBSERVATIONS (BEFORE FORMAL TESTING)

- **Gait observation:** Observe during entry to clinic — note shuffling pattern, heel clearance, step length, cadence; does FOG occur at the doorway? How does he manage the turn to sit down? Is there festination (involuntary acceleration)?
- **Postural stability:** Standing posture — degree of thoracic kyphosis and forward head position; sway; does he need to hold furniture?

Centre of mass position relative to base of support

- **Tone:** Cogwheel or leadpipe rigidity at wrist, elbow, and ankle; likely more severe than early PD; note whether it differs between "on" and "off" medication states if both can be observed
- **Dual-task observation:** Ask a simple question (name three cities) while patient walks — observe whether gait quality deteriorates markedly. A large dual-task cost is a strong independent predictor of falls in PD and confirms that the kitchen (a cognitively demanding environment) is inherently more dangerous for this patient
- **Facial expression and speech:** Degree of hypomimia; speech volume and clarity — note whether hypophonia affects his ability to call for help, which is a patient safety concern in the event of a floor fall
- **Lying-to-standing BP:** Before any standing-based assessment — measure supine BP after 5 minutes rest, then at 1 and 3 minutes post-standing; a drop of >20 mmHg systolic is clinically significant

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Test 1: Freezing of Gait Questionnaire (FOG-Q) + Timed 360° Turn Test

- **What it measures:** FOG-Q is a validated 6-item patient-reported outcome measure quantifying the frequency, duration, and severity of freezing episodes across different contexts (starting to walk, turning, going through doorways, distraction). The Timed 360° Turn Test objectively quantifies turning ability — the exact movement that caused both of Mr. Miller's falls — capturing time, step count, and FOG occurrence.
- **How to perform (360° Turn):**
  1. Ensure a safety spotter is present — stand close enough to catch without interfering
  2. Patient stands on a marked spot, feet shoulder-width apart

3. Instruct: "Turn a full circle to the right, then a full circle to the left. Walk at your usual pace."
4. Time each direction separately; count the number of steps to complete each 360°
5. Observe closely: does FOG occur? Does the patient pivot dangerously? Does festination develop? Is there step-crossing or near-fall?

- **Positive / Abnormal finding:** Normal = <3.5 seconds and <8 steps for a full 360°. More than 10–12 steps or a time >8 seconds is markedly abnormal. Any FOG during the turn is clinically significant — it replicates and confirms the mechanism of Mr. Miller's kitchen falls, providing objective evidence to guide my cueing strategy prescription and walking aid recommendation. The direction of greater difficulty (clockwise vs. anticlockwise) may also guide training emphasis.
- **Why I chose it:** Both falls occurred while turning — this test directly replicates the dangerous movement pattern. Confirming that FOG occurs during turning (rather than initiation from static, or doorways alone) allows me to prescribe targeted interventions: the wide U-turn strategy, visual cueing on the floor for turns, and training in turning within a simulated confined space. It also provides the baseline to measure whether my intervention reduces FOG frequency during this specific task.

Test 2: Pull Test (Postural Instability — MDS-UPDRS Item 3.12)

- **What it measures:** Automatic postural righting responses — whether Mr. Miller can recover his balance from an unexpected backward perturbation. At Hoehn & Yahr IV, I expect severely impaired postural reflexes. This test directly determines the severity of postural instability, guides walking

aid prescription, informs carer training in safe assistance, and is required for accurate UPDRS staging.

- **How to perform:**
  1. Perform with a second person present as a safety catcher at the front — do NOT perform alone in advanced PD
  2. Patient stands with feet shoulder-width apart; warn them: "I'm going to pull your shoulders backwards firmly — try to stay balanced, you may take steps"
  3. Stand directly behind patient, hands on both shoulders; deliver a firm, brisk backward pull
  4. Count corrective steps; note whether patient recovers or would fall without assistance
- **Positive / Abnormal finding:** At Hoehn & Yahr IV I anticipate 3–4 on the UPDRS pull test scale — meaning 3+ corrective steps or inability to recover without falling (would fall if not caught). This severity of postural instability confirms that Mr. Miller cannot safely recover from even a modest perturbation, which is precisely what happens during a FOG-triggered stumble. This finding justifies immediate walking aid prescription, carer training, and environmental modification as my first-line interventions — before any restorative exercise.
- **Why I chose it:** In advanced PD, postural instability is the primary driver of injurious falls and is the key factor that elevates Hoehn & Yahr stage. Quantifying it rigorously at this assessment does three things: (1) communicates urgency to the MDT; (2) guides whether restorative or compensatory approaches are the realistic primary treatment target; (3) provides essential information for carer training — knowing that Mr. Miller cannot recover from perturbation means his carer must understand how to assist a floor transfer safely.

Test 3: Dual-Task Walking Assessment (10-Metre Walk — Single vs. Dual Task)

- **What it measures:** The "Dual-Task Cost" (DTC) — the percentage decrease in gait performance when a concurrent cognitive task is added. In advanced PD, walking is no longer automatic; it requires significant attentional resources. When those resources are divided by any concurrent demand, gait deteriorates markedly, dramatically increasing fall risk. This directly models what happens in Mr. Miller's kitchen.
- **How to perform:**
  1. Measure a clear 10-metre walkway; mark start and stop lines
  2. Single task: ask patient to walk at comfortable pace end-to-end; record time and step count
  3. Dual task: repeat the walk while patient simultaneously counts backwards from 50 in 3s, or names alternating fruits and vegetables
  4. Calculate DTC = [(single-task speed − dual-task speed) / single-task speed] × 100%
  5. Also note whether FOG is triggered more easily during the dual-task condition
- **Positive / Abnormal finding:** DTC >20% is clinically significant; DTC >30% is associated with high falls risk. In advanced PD, a DTC of 30–60% is typical — meaning gait speed halves or worse when any cognitive demand is added. This directly explains why the kitchen — where Mr. Miller is planning tasks, responding to stimuli, and navigating spatially — is more dangerous than an open hallway. FOG occurring only during the dual-task condition but not single-task condition is a particularly high-risk finding.
- **Why I chose it:** Both falls occurred in a high-cognitive-demand environment (kitchen). If I confirm a large dual-task

cost, this changes my management: I will prioritise single-task strategies (stop and plan before moving; do not carry items while walking; sit down before reaching for objects in the kitchen) and introduce dual-task training as a progressive goal once single-task safety is established. Without quantifying DTC, I cannot communicate the mechanism of his kitchen falls clearly to the patient, carer, or MDT.

Outcome Measures

MDS-UPDRS Part III (Motor Examination): Administer separately during "on" and "off" medication states where safe and feasible — this quantifies the functional gap between these states, communicates the urgency of medication optimisation to the neurologist, and provides the most comprehensive motor baseline. Document time from last medication dose for each assessment.

Berg Balance Scale (BBS): 14-item standardised balance assessment scored 0–56. A score <45 indicates high fall risk; <36 indicates very high fall risk. At Hoehn & Yahr IV, I expect a substantially reduced score. BBS is appropriate at this severity because it includes tasks from sitting balance through tandem standing — the full range will reveal the severity of the deficit clearly. Administer during "on" period only for a valid and reproducible score.

Freezing of Gait Questionnaire (FOG-Q): 6-item scale (score 0–24; higher = more severe FOG). Used to track response to cueing strategies and exercise interventions over time and can be completed by patient or carer between appointments.

5 TREATMENT PLAN

INITIAL PHASE — WEEKS 1–4: RATIONALE

My primary goal in this phase is **falls prevention and immediate safety** — not restorative exercise. At Hoehn & Yahr IV, the capacity for neuroplastic restoration is substantially reduced compared to early PD. My approach is therefore principally compensatory: teaching strategies that bypass the deficient basal ganglia system using preserved cortical and cerebellar pathways, modifying the environment to reduce FOG triggers, and equipping Mr. Miller and his carer with effective real-world tools. All physiotherapy must be conducted during confirmed "on" periods and with a carer or therapist present throughout.

Exercise 1: Rhythmic Auditory Stimulation (RAS) Gait Training with Cueing

Clinical reasoning: In advanced PD, the basal ganglia can no longer internally generate the timing signals required for rhythmic gait. Rhythmic Auditory Stimulation provides an external timing signal via the auditory-motor coupling network — activating the supplementary motor cortex and cerebellum directly, which are relatively preserved in PD. Evidence demonstrates that RAS improves step length, gait velocity, and cadence in PD, and can significantly reduce FOG frequency. I apply this specifically to turning — the identified trigger — by using RAS to maintain rhythm through the turn rather than stopping.

How to perform: Set a metronome (phone app is sufficient) at approximately 10% above the patient's natural cadence to encourage longer step length. Instruct Mr. Miller to match each footfall to each beat. Begin with straight-line walking (supervised, clear space), then progressively introduce 90° turns (pause at corner; resume beat; step into the turn). Ensure one footfall per beat; do not rush. Practise in the specific environments where he freezes — recreate the corridor-to-kitchen transition if possible in the clinic.

- **Regression:** Seated marching to a slow metronome beat; therapist counts aloud as a substitute for the metronome;

practise on smooth wide floor only; straight-line gait only, no turns; gait belt applied by therapist throughout

- **Progression:** Introduce 90° then 180° turns to the metronome beat; carry a light object while walking (dual-task progression); practise on varied surfaces; transition patient to internal self-counting ("I say 1-2-1-2 silently to myself") to reduce device dependence; simulate kitchen layout in clinic space

Exercise 2: Safe Turning Strategy Training (Wide U-Turn + Visual Floor Cueing)

Clinical reasoning: The single-foot pivot turn — the instinctive way most people turn — is extremely dangerous in advanced PD: it requires full weight-bearing on one leg, crossing the other leg over (increasing instability), and is a direct FOG trigger in confined spaces. I replace this with the "wide U-turn" — multiple small steps in an arc — and use visual floor cues (bright tape or laser line) to provide an external visual target that activates the lateral premotor cortex and breaks or prevents FOG at the critical turning moment. This is a compensatory strategy that exploits preserved visual-motor pathways to bypass basal ganglia dysfunction.

How to perform (Wide U-Turn): Place a strip of brightly coloured tape on the floor as a visual target to step onto at the apex of the turn. Instruct: "When you want to turn, stop completely first. Look at the tape. Step onto it — do not cross your feet at any point. Keep stepping in a wide arc until you face the direction you want. Never pivot." Practise this movement slowly, with therapist walking alongside for support, until it becomes the default turning pattern. Also practise initiating forward walking from a freeze using the floor tape as a step-over target.

- **Regression:** Turn while supported by a kitchen counter (simulate with a plinth); therapist provides gentle pelvis guidance through the arc; use a very wide turning radius in

open space; turns only from standing, not while in motion; reduce to 90° turns before 180°

- **Progression:** Turn in progressively narrower spaces; turn while holding a light object (cup of water — high-fidelity kitchen simulation); turn from walking motion (not standing still); carer trained to shadow without touching; introduce at various medication states (safer relative "on" first)

SAFETY CONSIDERATIONS — ADVANCED PD (CRITICAL PRIORITY)

- **Never unsupervised at home during "off" periods:** Mr. Miller must have a carer present or immediately available whenever he is mobile during "off" periods; if alone, he should remain seated until medication takes effect
- **Walking aid review and prescription:** Evaluate for a laser-line walker (e.g. U-Step) — the built-in laser projects a bright line on the floor ahead of each step, providing a continuous visual cue that actively prevents FOG. A standard rollator may worsen FOG and should be reassessed; if changing aids, supervised practice is mandatory
- **Personal emergency alarm — prescribe immediately:** Mr. Miller has already had two falls at home; a wearable alarm that allows him to summon help from the floor is a non-negotiable safety device at this stage. Discuss with social worker if cost is a barrier
- **Kitchen environmental modification:** OT referral for: removal of rugs, widening turning radius by repositioning furniture, installation of grab rails at key points (stove, counter, doorway into kitchen); bright tape strips on the floor at known freezing points to serve as permanent visual cues; perching stool at the counter to allow seated task completion
- **Floor transfer training:** Teach Mr. Miller and his carer a safe method for getting up from the floor following a fall — rolling to one side, pushing to all fours, crawling to a stable piece of furniture, kneeling up, and then standing. Practise in a supervised, padded

environment. A carer who does not know how to assist a floor transfer may cause injury through well-meaning but incorrect manual handling

- **Bone health:** At 75, with advanced PD (reduced activity, poor vitamin D from reduced outdoor time) and a history of falls, osteoporosis is a significant co-risk. Flag to GP for DEXA scan and calcium/vitamin D assessment

PATIENT AND CARER EDUCATION — KEY MESSAGES

- **1. "Stop before you turn — always":** "The most dangerous moment is the turning pivot. From now on, your rule is: stop completely, look at your target, and step around in a wide arc. Never cross your feet. Never rush a turn. The kitchen is a high-risk zone — consider sitting down at the counter whenever you need to reach or cook, rather than standing and turning repeatedly."
- **2. Freezing is a brain signal problem — here is how to break it:** "When you freeze, your brain has temporarily lost its stepping rhythm. Do not try to force the step — this worsens the freeze. Instead: count aloud '1-2-1-2', march on the spot first, look down at a line on the floor and step over it, or rock your weight side to side until a step comes. These tricks use different brain pathways to restart movement."
- **3. Carer instruction — how to help safely:** "Walk alongside George, not behind. If he freezes, give him a rhythmic verbal count — do not pull him forward. If he falls, do not try to lift him; kneel beside him and guide him through the floor transfer sequence we've practised. Know how to activate the emergency alarm. And please tell us immediately if his 'off' periods are changing — that is a medical conversation that needs to happen urgently."
- **4. Medication timing is your safety window:** "Keep a diary of when 'on' periods start and stop — this will help the neurologist fine-tune your medication timing and doses. All walking and exercise should happen in your 'on' window. During 'off' periods,

stay seated unless you have assistance. This is not defeat — it is smart management of a predictable pattern."

PROGRESSION CRITERIA (MEASURABLE TARGETS TO ADVANCE)

- FOG-Q score decreases by ≥4 points from baseline
- Timed 360° turn completed without freezing episode using wide U-turn strategy (with metronome cueing)
- No further falls in the preceding 4 weeks
- Patient independently demonstrates wide U-turn and floor tape cueing strategy without verbal prompting
- Carer demonstrates correct FOG management technique and floor transfer assistance competently
- Personal emergency alarm in place and patient knows how to activate it

LONG-TERM PLAN — WEEKS 4–12+

- **Neurologist — urgent:** Report on-off fluctuation pattern with detailed diary; discuss deep brain stimulation (DBS) candidacy assessment if not already considered; discuss continuous dopaminergic delivery (apomorphine infusion or duodopa/levodopa intestinal gel) which can significantly reduce on-off fluctuations and FOG; medication timing adjustment based on diary data
- **Occupational therapy:** Comprehensive home assessment and major environmental modification; adaptive equipment for ADLs (dressing aids, bath seat, raised toilet frame); powered wheelchair assessment if walking capacity continues to decline; driving cessation counselling if not already addressed
- **Speech and Language Therapy:** Dysphagia screening and ongoing monitoring; LSVT LOUD for hypophonia if appropriate (improves communication and aspiration risk); communication aids if speech continues to deteriorate
- **Social work / palliative care:** Carer respite assessment and referral; community care package review; early advance care planning

discussions with Mr. Miller and family — conducted with compassion, supporting him to articulate his wishes while he has capacity

- **Maintenance physiotherapy:** Chair-based exercise programme for safe use during "off" periods; continued RAS-guided walking in "on" periods with carer; 3-monthly physiotherapy review with repeat UPDRS, Berg, and FOG-Q to track disease progression and adjust programme accordingly

🎓 Examiner Demonstration — What I Can Show In the Room
Technique 1 — Visual Floor Cueing to Break FOG: Place a strip of brightly coloured tape on the floor. Demonstrate to the examiner: when a patient freezes, stepping *onto* or *over* a visual floor target reliably breaks the freeze in many patients. Explain: "Visual cues activate the lateral premotor cortex and the visual-to-motor stepping system. These pathways are relatively preserved in PD — they are separate from the basal ganglia initiation circuit that has failed. A piece of tape on a kitchen floor can be genuinely life-changing for a patient with FOG. I would place permanent tape strips at every known freezing point in Mr. Miller's home."

Technique 2 — Safe Wide U-Turn Teaching: Demonstrate both turning strategies side by side — first the dangerous single-foot pivot (explain: "Full weight on one leg, crossing over, pivot — this is how a turn becomes a fall in advanced PD"). Then demonstrate the wide U-turn: multiple small steps in a clear arc, feet never crossing, wide base of support throughout. Invite the examiner to try both. Explain: "The wide U-turn keeps both feet under the base of support at all times. There is no single-limb stance phase during the turn. The trade-off is it takes longer and requires more space — which is why the kitchen layout modification is also essential. I would mark the wide turning arc on Mr. Miller's kitchen floor with tape so the path is always visually guided."

Neurological Physiotherapy OSCE Preparation · Cases 5 & 6 · Parkinson's Disease

Prepared for examination preparation. Clinical decisions must always reflect individual patient presentation and current best-evidence guidelines.

Key references: MDS-UPDRS (Goetz et al.), Hoehn & Yahr staging, LSVT BIG Protocol, Braak PD Staging, Freezing of Gait Questionnaire, MiniBESTest, Berg Balance Scale, Thaut et al. (RAS evidence), Parkinson's Physio Evidence Base (Physio for PD, 2024).